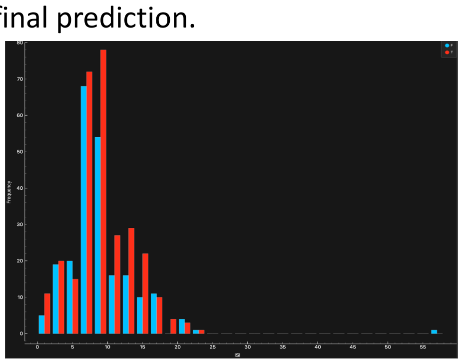
CSCU9M5 Assignment 1. Student Number 3346159

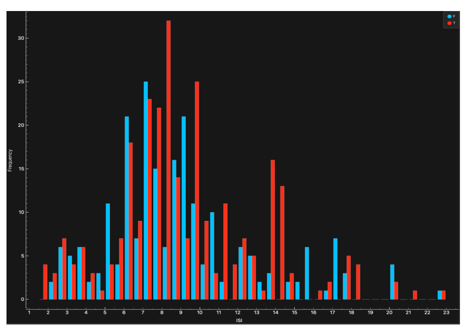
Project Methodology:

1. Loaded the data set into orange and inspected the variables.
2. Explored and understood the variables using the distributions widget to detect outliers and remove them.
3. Selected useful variables and removed low-value variables.
4. Created a 70/30 train-test split using data sampler.
5. Trained five different models using the 70% data (logistic regression, neural network, gradient boosting, random forest, kNN)
6. Tuned their hyper parameters and compared their F1 score as the main validation metric.
7. Selected gradient boosting as the final model as it had the highest score.
8. Used the confusion matrix to see how the model predicted.
9. Used the rank widget to understand which variable mostly influenced the prediction.

Data Preparation:

I decided to clean the ISI variable. It initially had a positively skewed graph with many outliers. The histogram on the left shows the variable before cleaning. It is wide and uneven. After removing several outliers, the histogram on the right shows much smoother and even distribution. This is done so that the model receives a consistent data and improves the final prediction.





Model training and hyper parameters:

I trained five different models Logistic Reasoning, Neural Network, gradient boosting, Random Forest, kNN. All these models are trained using a 70/30 train-test split. I used F1 score as the validation metric as it balances both precision and recall making it useful when dealing with class imbalance.
I tuned the hyper parameters for the top performing models to get a better score and ended up with gradient boosting as the best performing model.

Variables/ Data Understanding:

The variables used are:

Variables	type	Used in model
Month	Categorical (nominal)	Yes
FFMC, DMC, DC,ISI, TEMP, RH, WIND	Numeric (continuous)	Yes
area	Categorical (binary)	Yes (target)

All the environmental variables are treated as continuous numeric variables. The month variable is treated as nominal categorical. The target area variable is treated as binary categorical which shows whether the burned area exceeds 4%. The X, Y variables are not used as they don't really add a lot of information in the dataset. Most of the rain values are zero which is why it has been removed. The day variable don't necessarily help the model as we don't want to see weekly patterns which is why It has been removed as well.

Model	Hyper Parameters	Validation Metric (F1)
Logistic regression	C = 1; Regularisation type = L2	63.2 %
Neural network	Hidden layers = 100; Max iterations = 200	79.6 %
Gradient boosting	No of trees = 75; learning rate = 0.1	87.8 %
Random forest	No of trees = 100	85.8 %
kNN	No of neighbour= 10; Metric = Euclidean Weight =. uniform	82.4 %

Final Model and Results:

The final model was evaluated on the 30% test using confusion matrix. It correctly predicted : 74 large fire cases (true positives) and 56 non-large fire cases (true negatives) Which is around 88% correctly predicting the outcomes which matched with our F1 validation metric. The model made 9 false positives (predicting a fire when there is none) and 9 false negatives (failed to predict a fire).

Actual \ Predicted	F	T	Σ
F	56	9	65
T	9	74	83
Σ	65	83	148

Insight about data or model gained:

I used the rank widget to find out more information in how other variables affect the fire. The indices DC and DMC were most important followed by FFMC, temp, RH wind and ISI also contributing towards the final prediction. This suggests that the model mainly depends on the seasonal effects and dryness rather than any particular time or day which shows that the model is not biased towards any particular variable in the data. Month was mainly included to see when fires are more common during the year. Although, gradient boosting is good at predicting a large fire will occur, it doesn't really tell us why the prediction is made. This says that the model is accurate but not interpretable

			#	Info. gain	Gain ratio	Gini	χ²	ReliefF	FCBF	Gradi...sting
1	C	month	12	0.135	0.057	NA	35.337	0.048	NA	0.065
2	N	DC		0.118	0.059	0.072	25.128	0.044	0.086	0.384
3	N	DMC		0.091	0.046	0.058	39.490	0.066	0.000	0.276
4	N	FFMC		0.029	0.014	0.020	5.036	0.024	0.000	0.080
5	N	temp		0.027	0.013	0.018	13.699	0.014	0.000	0.064
6	N	RH		0.009	0.005	0.006	2.259	0.004	0.000	0.025
7	N	wind		0.007	0.003	0.005	1.021	0.010	0.000	0.023
8	N	ISI		0.006	0.003	0.004	2.788	0.029	0.000	0.034

References:

CRISP DM methodology : https://www.datascience-pm.com/crisp-dm-2/